

COUNT TO 15, THEN ANSWER ANY ESTIMATION QUESTION

How Many Ubers Are in San Francisco?

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1 Buy time, then count to 15

Tell the interviewer you're taking a moment, then breathe and count to 15 in your head. Do it even if you already know how you'd start. Use the count to decide what you'd ask first.

EXAMPLE

"Give me a few seconds to think this through."

When you hit 15, don't start solving. Ask your question.

2 Ask clarifying questions

For an estimation, ask one or two. For a design question, three to five. Pick the ones that change your answer the most. "Ubers" could mean cars on the road right now or every driver signed up, and those numbers are far apart.

EXAMPLE

"Do you mean Uber drivers on the road right now, or everyone signed up to drive? And Uber only, or Lyft too?"

Repeat their answer back before you move on: "Got it, Uber only, drivers on the road at 6pm on a weekday."

3 Break it apart on paper

Write the question at the top of the page and split it into layers before you do any math. Start with the biggest number you know and work down to the one they asked for. Drivers exist because riders exist, so riders go at the top.

EXAMPLE

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Uber drivers on the road in SF at 6pm
└ People in SF on a weekday: _____
  └ Share who take a rideshare: ___%
    └ Uber's share of those rides: ___%
      └ Rides in the 6pm hour: ___%
        └ Rides per driver per hour: ___
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Leave the blanks empty for now. Turn the page toward the interviewer so they can follow along.

4 Work it top down

Fill in the tree one layer at a time, out loud, and give a reason for every percentage. Round to numbers you can multiply in your head.

EXAMPLE

1. 826,000 people live in San Francisco. Add 300,000 who commute in or visit on a weekday.
 $826,000 + 300,000 \approx 1.1\text{M people in the city}$
2. Say 15% of them take one rideshare that day. Parking is expensive and the city is dense.
 $1.1\text{M} \times 15\% \approx 165,000 \text{ rides}$
3. Uber gets about 3 of every 4 of those rides.
 $165,000 \times 75\% \approx 125,000 \text{ Uber rides}$
4. 6 to 7pm is rush hour, so say 10% of the day's rides happen then.
 $125,000 \times 10\% = 12,500 \text{ rides at 6pm}$
5. A ride takes about 30 minutes counting pickup and traffic, so one driver does 2 an hour.
 $12,500 \div 2 \approx 6,250 \text{ Uber drivers}$

If they push back on a percentage, take their number and keep going.

5 Stress test your number

Check the answer against a fact you got a different way. Then name a corner case, the situation where your model breaks. On a design question, that's a power loss or a malfunction. On an estimation, it's the assumption that doesn't hold every day.

EXAMPLE

"The city's transportation authority estimated about 170,000 Uber and Lyft trips a day, so my 165,000 holds up. They also estimated about 5,700 Uber and Lyft cars at the weekday peak, and I have 6,250 for Uber alone. So my rides per hour is too low. If a ride takes 20 minutes, a driver does 3 an hour, and I get about 4,200 Uber drivers, which puts both apps near 5,600. Corner case: that count is from 2016, before the pandemic and before robotaxis. My answer is 4,000 to 6,000 Uber drivers on a weekday evening."

Close on the range and name the assumption that would move it most. Here, that's rides per driver per hour.

Practice on these

Set a 5-minute timer and run all five steps out loud, count included. Record at least two, then play them back and flag any number you said without a reason.

- ☐ How many Waymo rides happen in San Francisco each day?
- ☐ How many Chick-fil-A sandwiches get sold in the US on a Saturday?
- ☐ How many DoorDash orders go out in New York City on a Friday night?
- ☐ How many Amazon packages get delivered in Atlanta in a day?
- ☐ How many people fly out of the Atlanta airport today?

Score every answer

Use this checklist after each practice run:

- ☐ Did I say "give me a few seconds" and count to 15?
- ☐ Did I ask one or two clarifying questions and repeat the scope back?
- ☐ Did I sketch the layers on paper before any math?
- ☐ Did I give a reason for every percentage?
- ☐ Did I say each line of math out loud?
- ☐ Did I check my number against a second fact and name a corner case?